Session 1 · Thursday 27 August 2026 · Groups of 3–4. Fifteen minutes. No answers yet.

Three things someone wants built

Each of these is a real specification that a real funder has paid for. You have the specification. You do not yet have any of the course.

For each one, write down: what would you have to KNOW, and what would you have to BUILD? Then rank the three by difficulty, and be ready to defend the ranking — the ranking is the part we will argue about.

A · An arsenic sensor

> *A bacterium that turns visibly red when the well water it is in exceeds 10 µg/L arsenic.*

Context: 10 µg/L is the WHO drinking-water limit. Tens of millions of people in Bangladesh and West Bengal draw water from wells that may or may not exceed it, and the existing field test is a chemical kit costing a few dollars per well. The output has to be readable by eye, by someone who is not a scientist, in a village.

Know:

Build:

B · A tumour-selective cell therapy

> *A human T cell that kills a tumour cell and spares a healthy cell of the same tissue.*

Context: engineered T cells work spectacularly in some blood cancers. In solid tumours they mostly do not, and the failures include patients killed by the therapy attacking healthy tissue that shares a marker with the tumour. The tumour is not a different organism; it is the patient's own cells with a different pattern of expression.

Know:

Build:

C · A cereal that fixes its own nitrogen

> *A cereal crop that fixes its own nitrogen and needs no synthetic fertiliser.*

Context: legumes host bacteria that convert atmospheric N_2 to ammonia; cereals — wheat, rice, maize, which is most of the calories humans eat — do not. Making synthetic fertiliser consumes on the order of 1–2% of world energy supply, and the run-off is a major cause of coastal dead zones. The enzyme that does the chemistry, nitrogenase, is irreversibly poisoned by oxygen and is expensive to run.

Know:

Build:

Ranking

Hardest ► ____________ ► ____________ ► ____________ ◄ easiest

Why? One sentence. The reason matters more than the order.